

# Trained PING streams sequential digits without retraining

exp048 · 8 June 2026

## Abstract

A PING network trained one digit at a time on 200 ms trials can classify a *stream* of digits at a fraction of that time each, with no retraining: only the readout’s integration window changes. At  $\tau = 50$  ms ( $\approx 2$  gamma cycles) 4 of the 5 sequential MNIST digits classify correctly, the readout flipping within one cycle of each transition. A 2-D sweep over ( $\tau$ , input rate) reveals a sub-cycle failure floor below  $\approx 15$  ms and a broad high-accuracy plateau above it. An extended 200 ms rate slice remains at chance through 0.5 Hz and becomes clearly informative by 2 Hz, locating the encoder’s low-rate information floor.

## Method

Canonical exp025 PING baselines (1,024 excitatory + 256 inhibitory cells, trained one MNIST digit per 200 ms trial, three seeds 42, 43, 44). Everything here is **inference-only** at the trained  $\Delta t = 0.10$  ms; the weights are never updated. The 2-D sweep averages over the three seeds; the single-stream demos use seed 42.

A stream of digits, each shown for  $\tau$  ms, is classified in one forward pass. The *only* change from training is a sliding readout window:

1. **Encode** each digit as a Poisson spike train over  $\tau$  ms across 784 input channels, one per pixel, firing rate proportional to pixel intensity at the chosen input rate.
2. **Concatenate** the per-digit trains into one input stream. Transitions are *hard switches*: the Poisson rate flips instantaneously at  $t = k\tau$ , with no blending.
3. **One forward pass** of the trained network over the whole stream (no retraining), at the trained  $\Delta t$ .
4. **Integrate evidence** in a non-spiking output leaky-integrator, one unit per class:

$$v_{\text{out}}(t) = \beta_{\text{out}} v_{\text{out}}(t-1) + \frac{1 - \beta_{\text{out}}}{\Delta t} \mathbf{s}^E(t-1) W_{\text{out}}.$$

5. **Read a sliding window.** Average  $v_{\text{out}}$  over the trailing  $\tau$ -window and softmax it:

$$\text{logits}(t) = \frac{\Delta t}{\tau} \sum_{u=t-w+1}^t v_{\text{out}}(u), \quad p(\text{class}, t) = \text{softmax}(\text{logits}(t)),$$

At training the average ran over the *whole* trial; the trailing window is the single change.

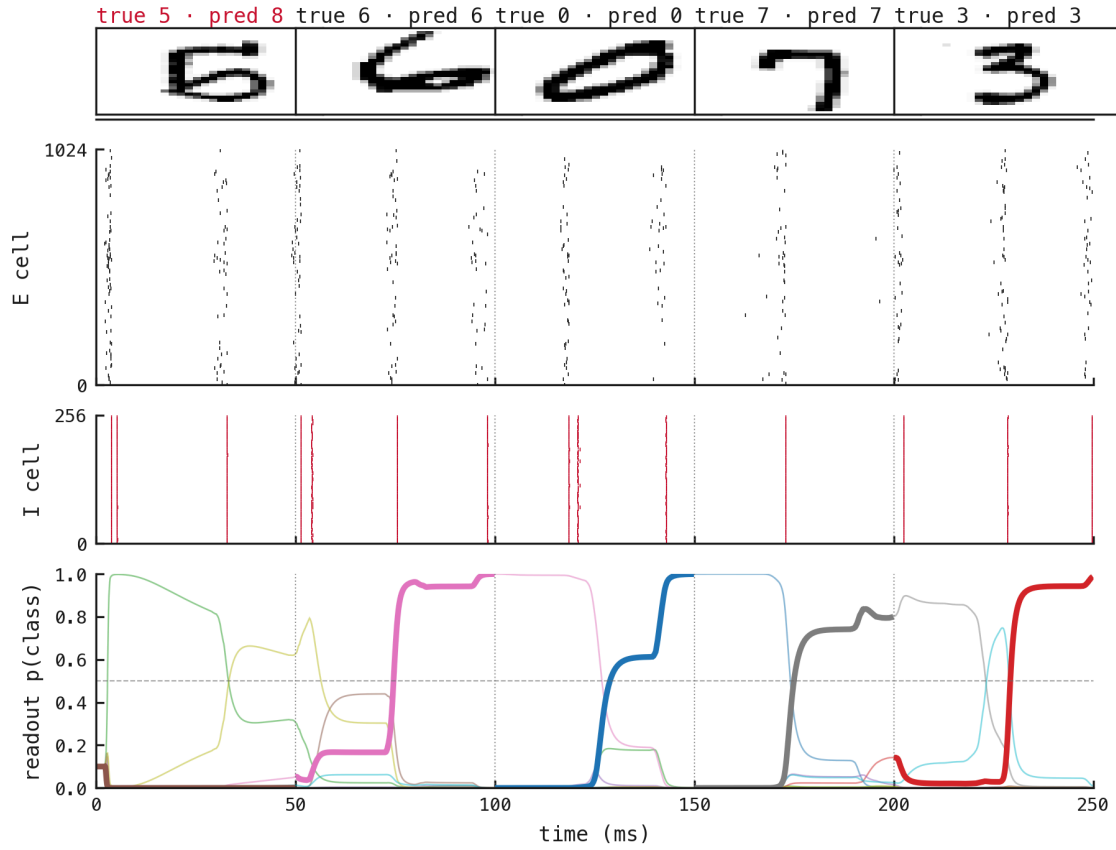
6. **Predict per segment** as  $\arg \max_c p(\text{class} = c, t)$ , read at the end of each digit’s  $\tau$ -window.

Here  $\mathbf{s}^E(t)$  is the excitatory spike vector,  $W_{\text{out}}$  the trained readout (unchanged), and  $\beta_{\text{out}} = \exp(-\Delta t / \tau_{\text{out}})$  the output leak with  $\tau_{\text{out}} = 2$  ms. The probability trace  $p(\text{class}, t)$ , the network’s online confidence in each digit, is what the figures plot.

The grid sweeps  $\tau$  over 10, 15, 25, 40, 50, 75, 100, 200 ms and input rate over 5, 10, 25, 50, 100, 200 Hz per channel, giving  $8 \times 6$  cells, each 40 streams  $\times$  10 digits  $\times$  3 seeds (1200 segments per cell). It extends down to  $\tau = 10$  ms ( $\approx 0.36$  of a gamma cycle) to resolve the sub-cycle regime.

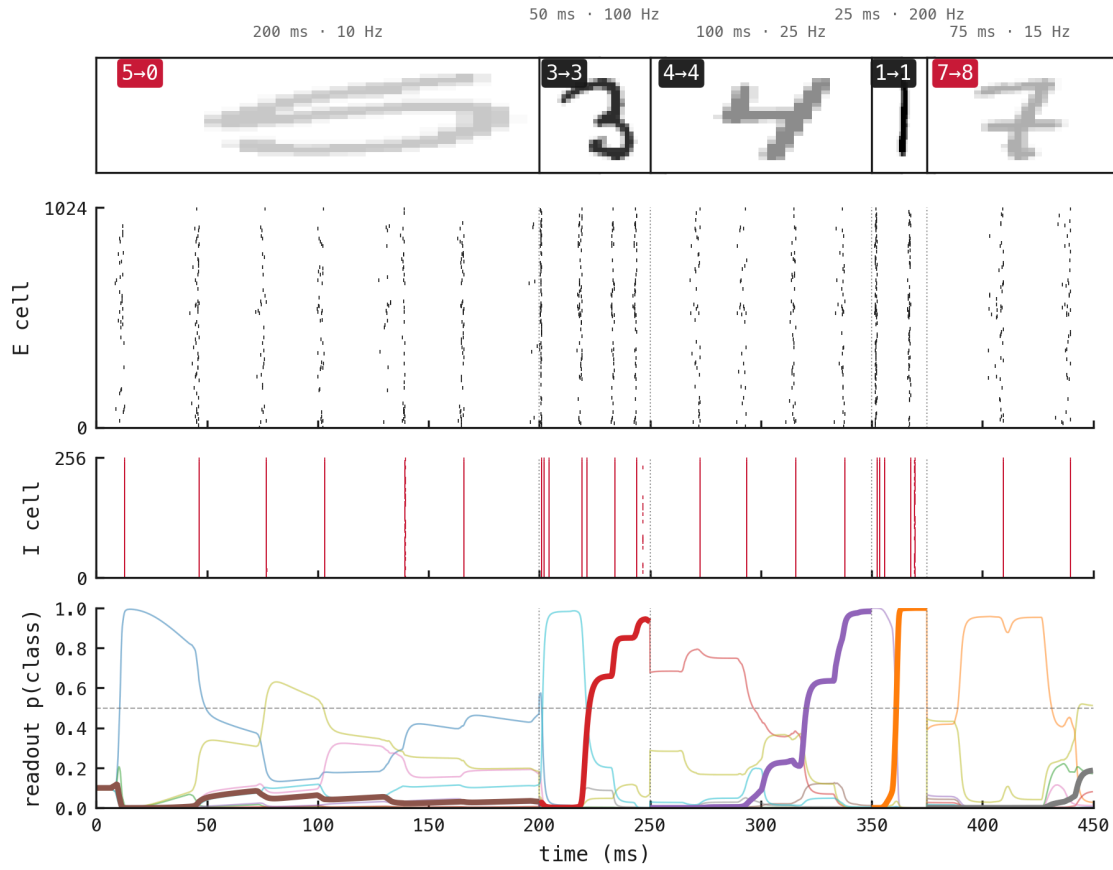
To resolve the encoding-rate floor below the grid, an additional 200 ms slice sweeps rates from 0.01 to 3 Hz. Each cell contains 10 streams of 10 digits across the same three trained seeds. The published 5–200 Hz points are the matching 200 ms row of the grid.

## Results



replot

Figure 1: Five sequential digits (5, 6, 0, 7, 3) at  $\tau = 50$  ms each ( $\approx 2$  gamma cycles per digit, 250 ms total, 25 Hz input). The gamma cadence ( $\approx 28$  ms) is preserved across the stream; the readout flips to the new digit within one cycle of each transition and reaches near 1.0 by the segment's end. **4/5 correct** (the leading 5 is misread). The figure makes four things visible at once: sparse E firing, the I-burst clock, class-tracking readout, and re-identification within  $\approx 1$  cycle of each switch.



exp048-replot

Figure 2: A harder test: each digit gets its *own* duration and input rate within one stream (durations 25–200 ms, rates 10–200 Hz). Thumbnail opacity  $\propto$  input rate (faint = weak drive, bold = strong); the sliding window uses each segment's own  $\tau$ , so every digit's prediction respects its presentation window. **3/5 correct** here: the two errors fall on the weakest-drive segments (a 5 at 10 Hz and a 7 at 15 Hz), the two lowest input rates in the stream.

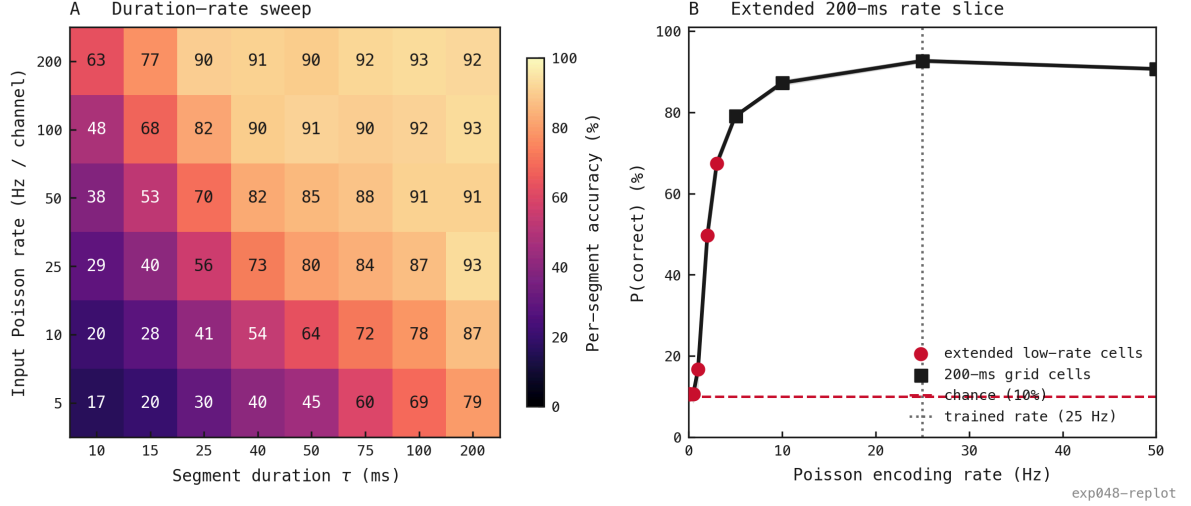


Figure 3: Temporal and encoding-rate limits of the frozen PING classifier. **(A)** Per-segment accuracy over the full duration  $\times$  rate grid (3 seeds  $\times$  1200 segments per cell). The sub-cycle regime fails even under strong drive, while above one cycle duration and rate trade off along diagonal iso-accuracy contours. **(B)** The 200 ms row extended below 5 Hz on a linear rate axis. Circles are the new low-rate cells; squares come from panel A. The dashed line marks ten-class chance and the dotted line the 25 Hz training rate. Accuracy remains on its empty-input floor through 0.5 Hz, is clearly informative by 2 Hz, and reaches 79.1% at 5 Hz.

The extended slice distinguishes a nonviable encoder regime from ordinary classification errors under weak evidence. In the variable-condition stream, the failed 5 received 10 Hz for 200 ms, yet that condition reaches 87.3% across the population. Its error is therefore natural trial-level variation, not evidence that 10 Hz is intrinsically too low. The failed 7 at 15 Hz and 75 ms is likewise above the empty-input rate floor, although its shorter window supplies less total evidence. Rates below 0.5 Hz are not useful operating points; 2 Hz is the lowest clearly informative tested rate and 5 Hz is a practical lower bound for future sweeps.